

STATATHON 2025

TITLE PAGE

- **Problem Statement ID:** 4
- **Problem Statement Title:** AI Enhanced Application for Automated Data Preparation, Estimation, And Report Writing.
- **PS Category-Software:** Data Processing and Analysis.
- **Team ID:** 7811
- **Team Name (Registered on portal):** InnoStat_77

IDEA / APPROACH

ASDP (AI Survey Data Processor) - [AI-Augmented Platform]



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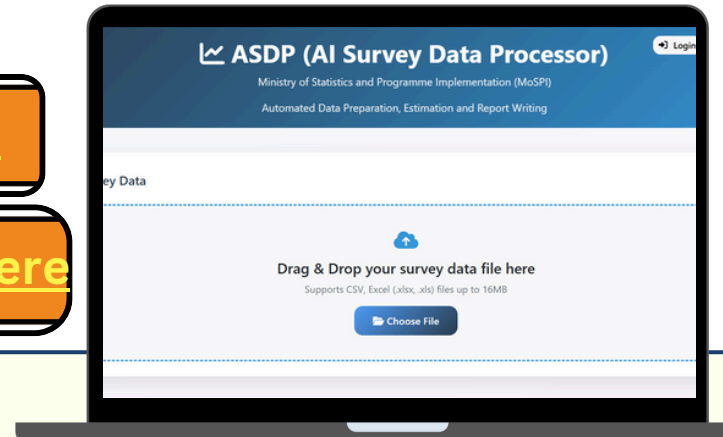
Detailed explanation of the proposed solution :

- The **platform** takes in **raw survey** files like CSV or Excel and maps the schema for smooth processing.
- It cleans the data using **imputation**, **outlier checks**, and **rule validation** to ensure accuracy.
- Survey weights are applied to generate reliable estimates with **margins of error**.
- Finally, it produces **ready-to-use PDF or HTML reports** through a simple, **user-friendly** interface.

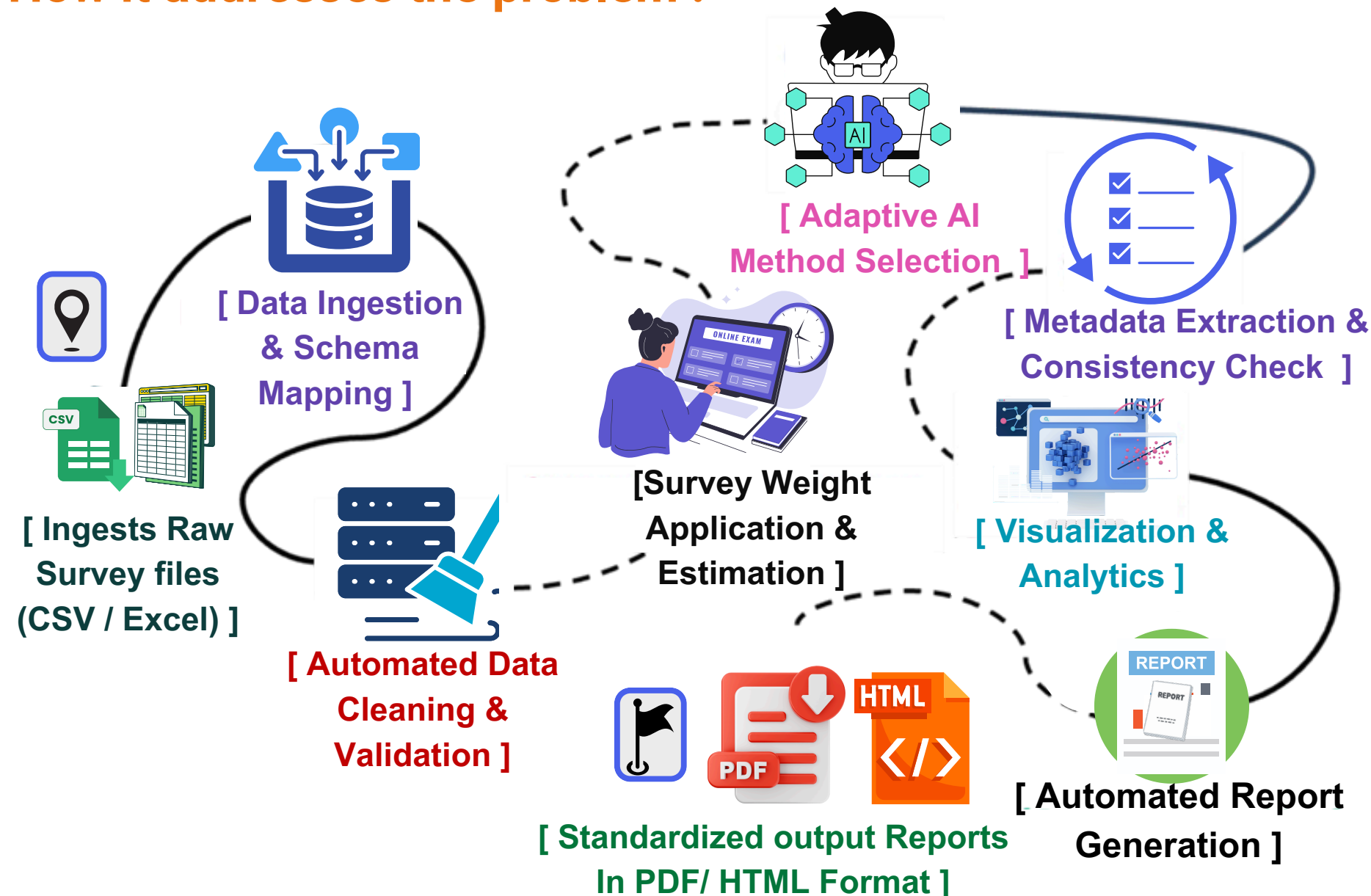
Seamless integration of
ASDP with MOE Website

[Prototype Link : Click here](#)

[More Screenshot : Click Here](#)



How it addresses the problem :



Innovation And Uniqueness :

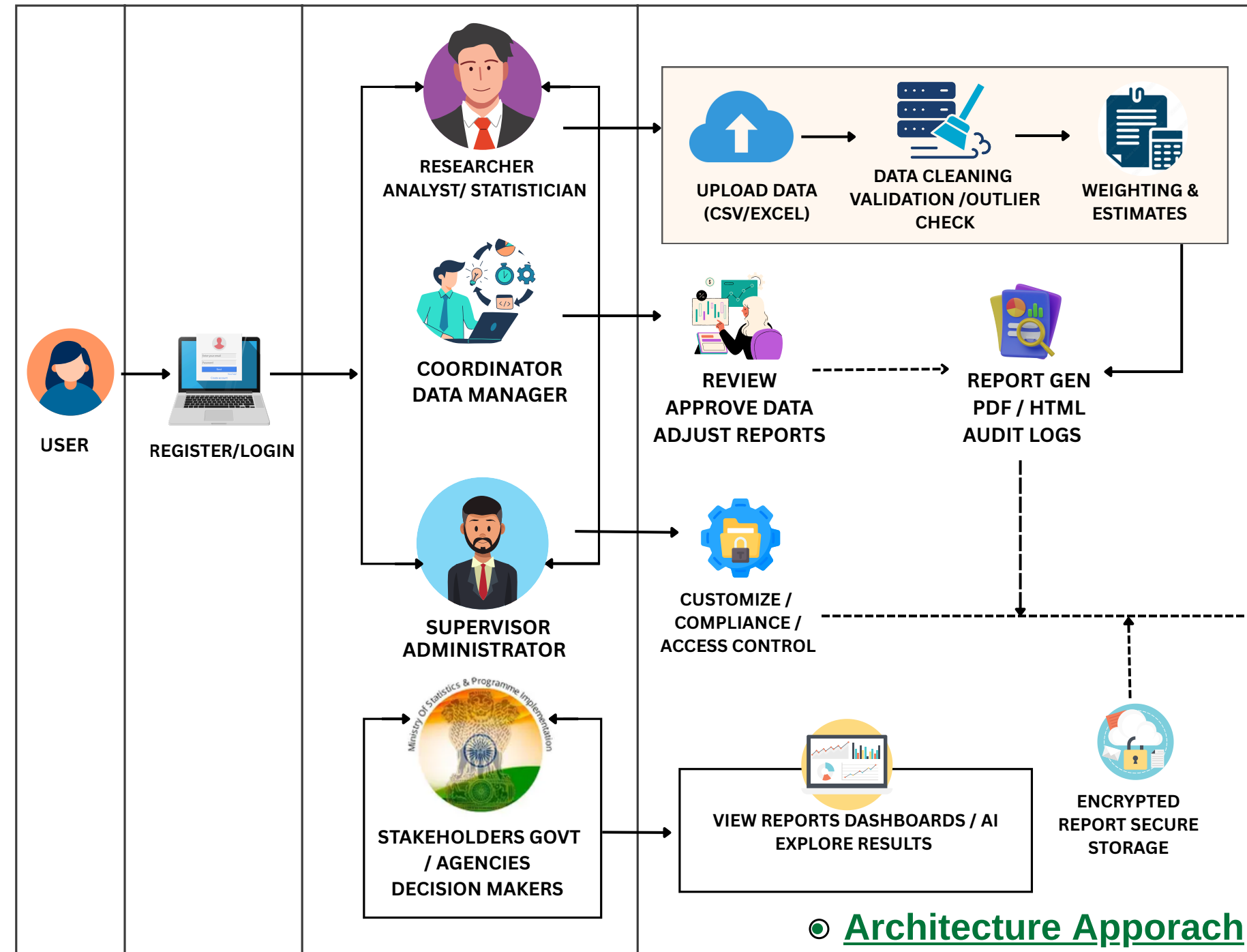
- **End-to-End AI Platform:** Unified pipeline for cleaning, weighting, estimation, and reporting.
- **AI Processing:** Smart imputation, anomaly detection, and validation with explainability.
- **Report Q&A (LLM + RAG):** Ask questions directly from generated reports.
- **Automated Workflows (Airflow):** Orchestrated tasks with scalability and continuous learning.
- **Secure & Open-Source:** Platform-independent with strong security and open tools only.
- **Real-Time Analytics:** Instant anomaly and trend detection during processing.
- **Multilingual + Voice:** Reports and interface in English, Hindi, and regional languages.
- **Low-Code Config:** Simple setup for non-technical users.
- **Audit Trails:** Workflow logs for transparency and reproducibility.
- **Survey Design Assist:** AI suggests improved survey design for future rounds.

TECHNICAL APPROACH



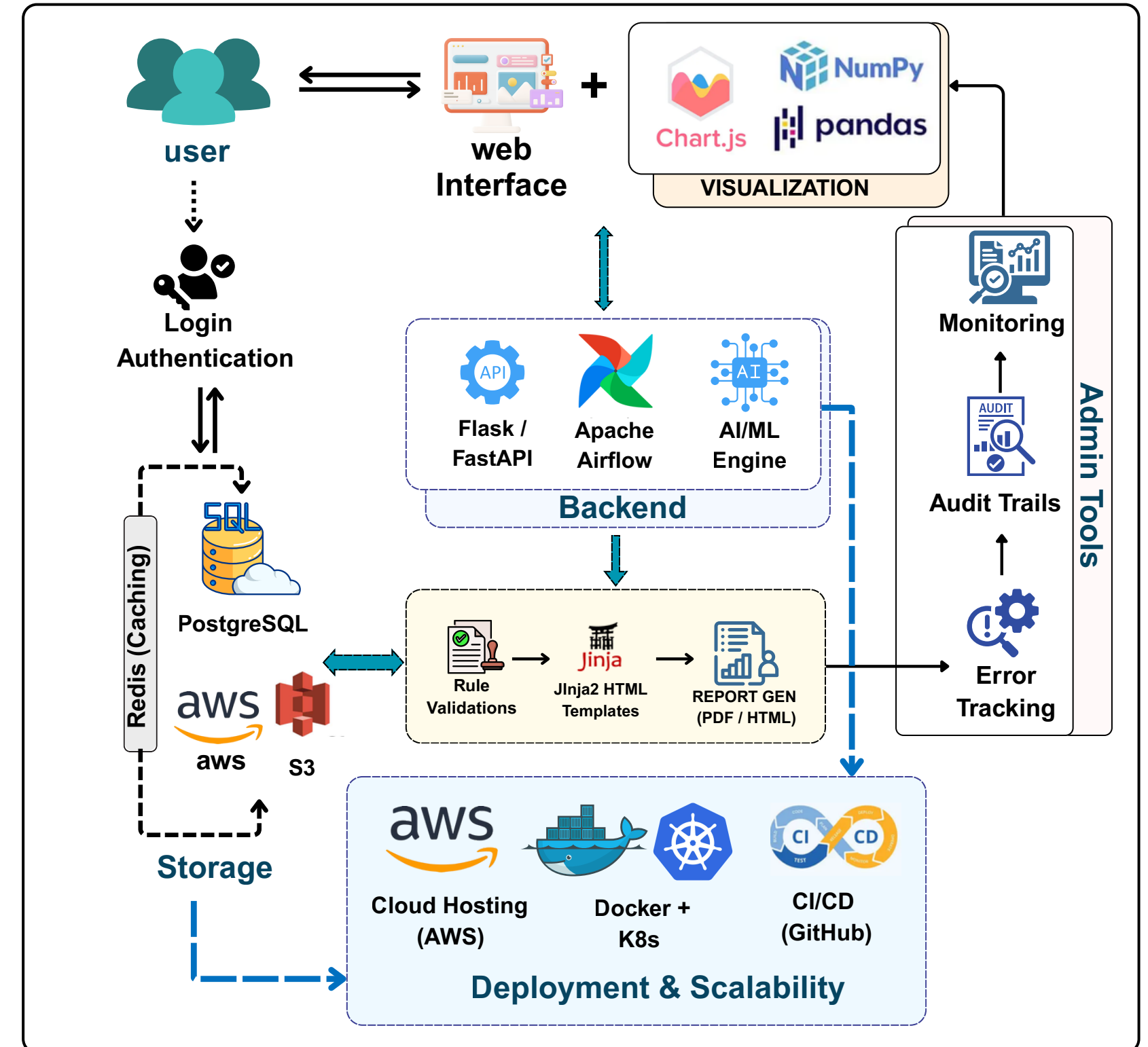
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Flow Process Diagram



- ◉ **Users:** upload survey data, view dashboards, and query via AI assistant.
- ◉ **Coordinators:** review processed data, adjust, and approve reports.
- ◉ **Administrators:** customize formats, ensure compliance, and manage access.
- ◉ **Stakeholders:** receive encrypted reports and explore insights.

Architecture Diagram:


[Prototype Link : Click here](#)
[Project Files : Click here](#)



FEASIBILITY AND VIABILITY

◆ Feasibility of the Idea

- **Growing Demand:** Increasing need for automation in data processing for large-scale surveys.
- **Efficiency Gains:** Reduces preprocessing time by 60–70% compared to manual workflows.
- **Cost Reduction:** Lower operational costs via automated pipelines and reusable modules.
- **Scalability:** Easily adaptable across national surveys, research institutions, and state-level agencies..

◆ Potential Challenges & Risks

- **Data Quality Issues:** Missing values, outliers, and inconsistent schema across datasets.
- **Implementation Costs:** Initial setup of cloud infra, AI models, and integration tools.
- **Skill Gap:** Need for training staff to operate AI-enabled platforms.
- **Security Concerns:** Handling sensitive government survey datasets securely.

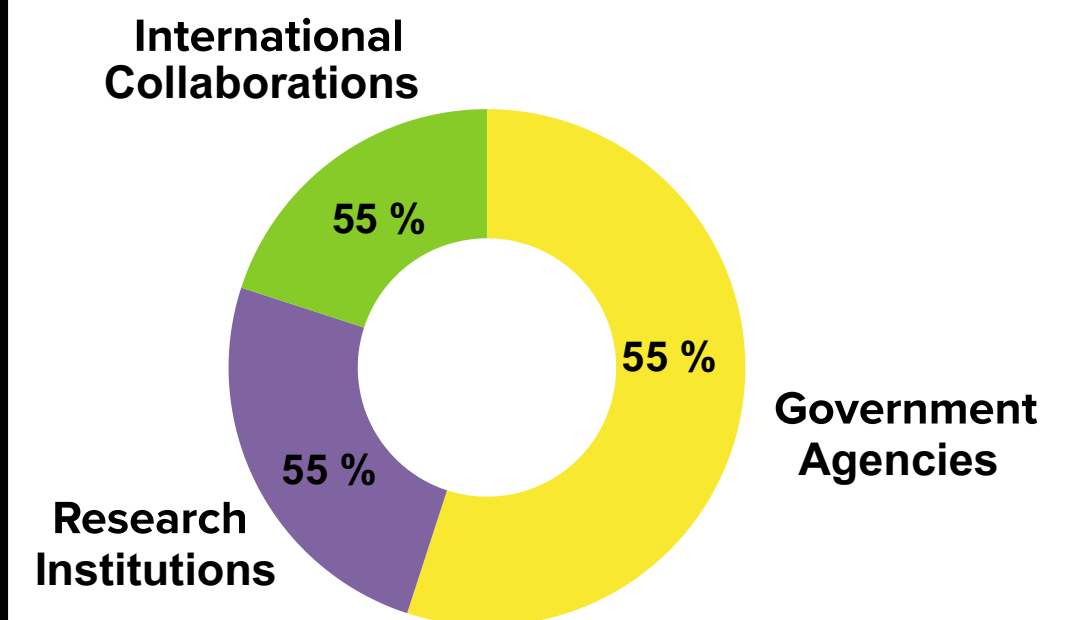
◆ Strategies to Overcome Challenges

- **Data Quality Handling:** AI-driven imputation, validation rules, and audit trails.
- **Cost Flexibility:** Cloud-based scalable architecture to reduce infrastructure costs.
- **User Support:** Training and user-friendly interface to simplify adoption.
- **Robust Security:** Encryption, role-based access, and compliance with data standards.

◆ Long-Term Market Viability

- **Efficiency Growth:** Automating survey preparation saves resources and reduces turnaround.
- **Sustainable Impact:** Scalable solution supporting MoSPI's digital initiatives and reproducibility.
- **User Accessibility:** Configurable interface enables wide adoption by statistical staff.

◆ Visuals



IMPACT AND BENEFITS



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Impact and Benefits for Users (Statisticians, Analysts, Policymakers):

- **24/7 Data Access** – Upload survey files anytime and get instant cleaned, weighted, and analyzed outputs.
- **Faster Results** – AI automation reduces processing from weeks to hours.
- **Easy Customization** – Choose cleaning methods, outlier rules, and weights without coding.
- **Better Data Quality** – Automatic error detection and correction.
- **Transparent Outputs** – All steps logged for trust and reproducibility.

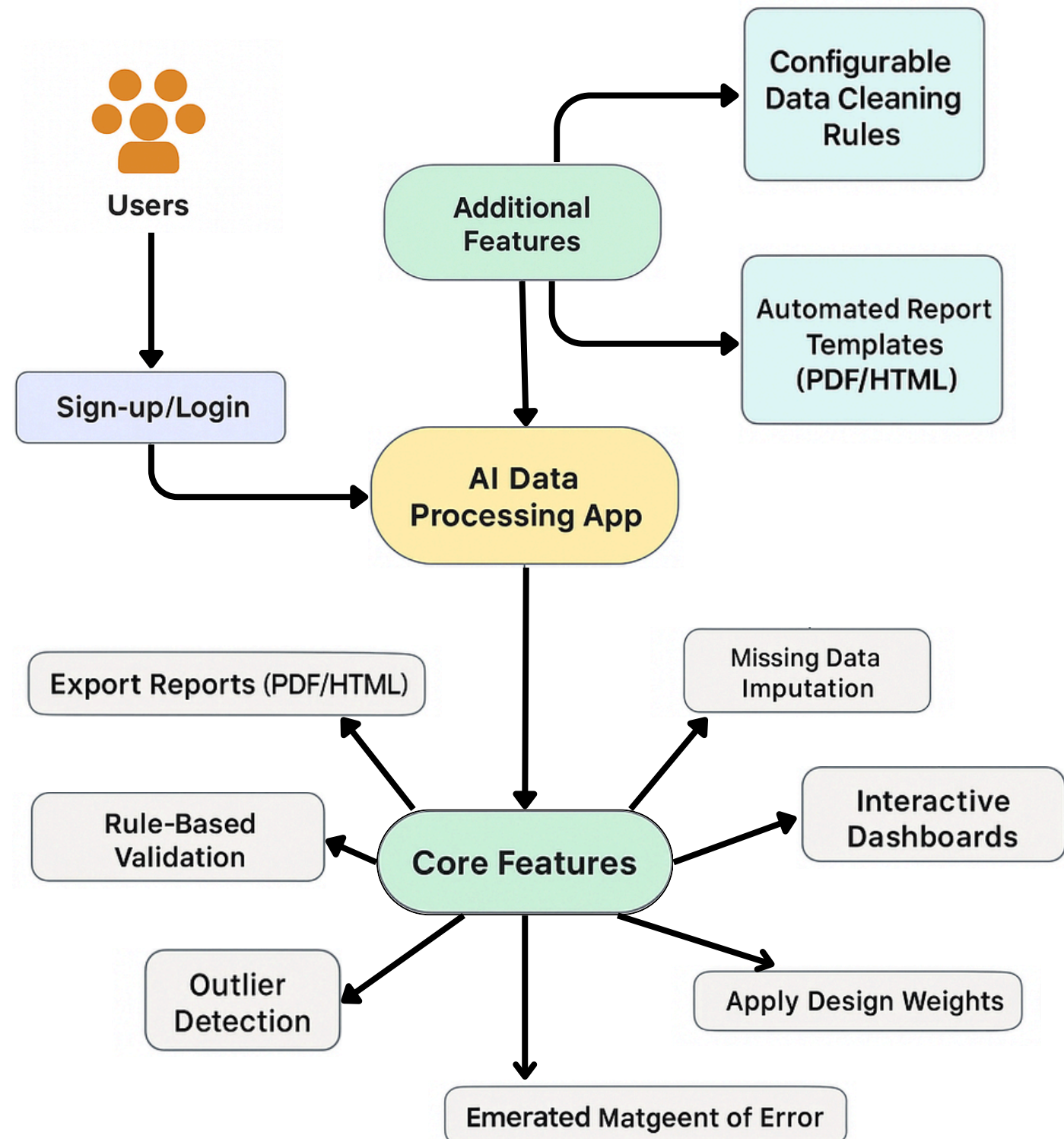
Impact and Benefits for Agencies (MoSPI & Other Statistical Bodies):

- **Less Workload** – Automates repetitive data prep, freeing staff for deeper analysis.
- **Cost Savings** – Cuts manpower and operational costs.
- **Scalable** – Handles large datasets and multiple surveys quickly.
- **Standardized Reports** – Generates official-ready PDF/HTML outputs.

Business Model :-

- SaaS Subscriptions – Basic, Pro, and Enterprise plans (monthly/annual) based on datasets, storage, and AI features.
- Pay-Per-Use – Flexible pricing for occasional users like NGOs or consultants.
- Custom Enterprise Solutions – Tailored data pipelines and on-premise deployments for privacy-focused clients.
- Training & Support – Paid onboarding, workshops, and priority technical assistance.

Representation of Project:-



[Youtube Link : Click here](#)

RESEARCH AND REFERENCES



Books Reference

- [1] Harrington, P. (2012). Bad Data Handbook: Cleaning Up the Data So You Can Get Back to Work. O'Reilly Media
- [2] Dasu, T., & Johnson, T. (2003). Exploratory Data Mining and Data Cleaning. Wiley-Interscience.
- [3] Berman, J. (2013). Principles of Data Quality. Springer.

Videos Reference

- [1] <https://www.coursera.org/courses?query=data%20cleaning>
- [2] https://youtu.be/bDhvCp3_IYw?si=XoYXB0XNkwDvZG1V
- [3] https://youtu.be/oRM9X8nGQC0?si=LfmeoE9sq_n6QQW7

General References

- [1] A. A. M. Aljuffri, C. Reinbrecht, S. Hamdioui, and M. Taouil, "Impact of data pre-processing techniques on deep learning based power attacks," in 2021 16th International Conference on Design & Technology of Integrated Systems in Nanoscale Era (DTIS), 2021, pp., doi: 10.1109/DTIS53253.2021.9505051.
- [2] Faizal, B., Abraham, S., & Thomas, S. (2024, March). Automated Business Report Summarization Using Transformer Model. In 2024 10th International Conference on Advanced Computing and Communication Systems (ICACCS) (Vol. 1, pp. 254-258). IEEE.
- [3] Kim, D. (2017). An Algorithmic Approach to Personalized and Interactive News Generation